

# Handling Header & Item Data in ABAP

Using FOR ALL ENTRIES Effectively

# Problem Statement:

## Requirement:

- Fetch Order Header data.
- Fetch corresponding Item data.
- Combine both into a single output.

## Challenge:

- Data exists in **two different tables**.

Code:

```
1  * &-----
2  * & Report ZP34_CR_FOR_ALL_ENTRIES
3  * &-----
4  * &
5  * &-----
6  REPORT zp34_cr_for_all_entries.
7
8  DATA : lv_ono TYPE zordernum_pl.
9  DATA : lv_flag TYPE boolean.
10
11 TYPES : BEGIN OF lstr_tab1,
12         order_number TYPE zordernum_pl,
13         order_date    TYPE zorderdate_pl,
14         payment_mode  TYPE zpayment_pl,
15         currency      TYPE zcurrency_pl,
16     END OF lstr_tab1.
17
18 DATA : lit_tab1 TYPE TABLE OF lstr_tab1.
19 DATA : lwa_tab1 TYPE lstr_tab1.
20
21 TYPES : BEGIN OF lstr_tab2,
22         order_number  TYPE zordernum_pl,
23         order_itm_num TYPE zorderitmno_pl,
24         item_cost     TYPE zitemcost_pl,
25     END OF lstr_tab2.
26
```

```

26
27 DATA : lit_tab2 TYPE TABLE OF lstr_tab2.
28 DATA : lwa_tab2 TYPE lstr_tab2.
29
30 □ TYPES : BEGIN OF lfinal,
31     order_number    TYPE zordernum_pl,
32     order_date      TYPE zorderdate_pl,
33     payment_mode    TYPE zpayment_pl,
34     currency        TYPE zcurrency_pl,
35     order_itm_num   TYPE zorderitmno_pl,
36     item_cost       TYPE zitemcost_pl,
37     END OF lfinal.
38
39 DATA : lit_final TYPE TABLE OF lfinal.
40 DATA : lwa_final TYPE lfinal.
41
42 SELECT-OPTIONS : s_ono FOR lv_ono.
43
44 SELECT order_number order_date payment_mode currency
45 FROM zohtable_pl
46 INTO TABLE lit_tab1
47 WHERE order_number IN s_ono
48 ORDER BY order_number.
49

```

```

44 SELECT order_number order_date payment_mode currency
45 FROM zohtable_pl
46 INTO TABLE lit_tab1
47 WHERE order_number IN s_ono
48 ORDER BY order_number.
49
50 IF lit_tab1 IS NOT INITIAL." NOT INITIAL - not null
51     SELECT order_number order_itm_num item_cost
52     FROM zoitmtable_pl
53     INTO TABLE lit_tab2
54     FOR ALL ENTRIES IN lit_tab1
55     WHERE order_number = lit_tab1-order_number.
56 ENDIF.
57

```

```

58  LOOP AT lit_tab1 INTO lwa_tab1.
59  LOOP AT lit_tab2 INTO lwa_tab2 WHERE order_number = lwa_tab1-order_number.
60      lwa_final-order_number = lwa_tab1-order_number.
61      lwa_final-order_date = lwa_tab1-order_date.
62      lwa_final-payment_mode = lwa_tab1-payment_mode.
63      lwa_final-currency = lwa_tab1-currency.
64      lwa_final-order_itm_num = lwa_tab2-order_itm_num.
65      lwa_final-item_cost = lwa_tab2-item_cost.
66      APPEND lwa_final TO lit_final.
67      CLEAR : lwa_final.
68      lv_flag = 'X'.
69  ENDLOOP.
70  IF lv_flag = ' '.
71      lwa_final-order_number = lwa_tab1-order_number.
72      lwa_final-order_date = lwa_tab1-order_date.
73      lwa_final-payment_mode = lwa_tab1-payment_mode.
74      lwa_final-currency = lwa_tab1-currency.
75      APPEND lwa_final TO lit_final.
76      CLEAR : lwa_final.
77  ENDIF.
78      CLEAR : lv_flag.
79  ENDLOOP.
80
81  LOOP AT lit_final INTO lwa_final.
82      WRITE : / lwa_final-order_number, lwa_final-order_itm_num, lwa_final-order_date,
83              lwa_final-payment_mode, lwa_final-item_cost, lwa_final-currency.
84  ENDLOOP.

```

# Step-by-Step Explanation of Code:

## 1) Header Data Fetch:

Code :

```
SELECT order_number order_date payment_mode currency  
FROM zohtable_p1  
INTO TABLE lit_tab1  
WHERE order_number IN s_ono.
```

Result :

- All matching header records stored in internal table.

## 2) FOR ALL ENTRIES:

Code:

```
IF lit_tab1 IS NOT INITIAL.  
  SELECT order_number order_itm_num item_cost  
  FROM zoitmtable_p1  
  INTO TABLE lit_tab2  
  FOR ALL ENTRIES IN lit_tab1  
  WHERE order_number = lit_tab1-order_number.  
ENDIF.
```

Key Concept:

- Fetches item data for all header entries.

Critical Rule:

- If lit\_tab1 is empty → full table fetch (performance issue).



# Why Not SELECT Inside LOOP?

Bad Practice:

- SELECT inside LOOP → multiple DB hits.

Better Approach:

- FOR ALL ENTRIES → single DB hit.

→ Performance optimized design.

### 3) Data Merging Logic:

Code :

```
LOOP AT lit_tab1 INTO lwa_tab1.  
  LOOP AT lit_tab2 INTO lwa_tab2  
    WHERE order_number = lwa_tab1-order_number.
```

Concept:

- Manual join in ABAP layer.
- Combines header + item data.

## 4) Building Final Structure:

Code :

```
APPEND lwa_final TO lit_final.
```

Why final table?

- Unified dataset
- Ready for display / ALV

Real-time:

- Used before ALV reports or output formatting.

## 5) Handling Missing Items:

Code :

```
IF lv_flag = ''.
```

Insight:

- Handles cases where:
  - Header exists
  - No item records found

Ensures:

- No data loss.
- Complete reporting.

## 6) Output Display:

Code :

```
LOOP AT lit_final INTO lwa_final.  
WRITE : / ...  
ENDLOOP.
```

Final step:

- Displays merged dataset.

## Selection Screen:

***To display the classical reports for order header and order item table***



order number  to  

## Output Screen:

***To display the classical reports for order header and order item table***

|                                                                        |    |            |   |          |     |
|------------------------------------------------------------------------|----|------------|---|----------|-----|
| To display the classical reports for order header and order item table |    |            |   |          |     |
| 0000000001                                                             | 10 | 01.09.2025 | C | 100,00   | INR |
| 0000000001                                                             | 20 | 01.09.2025 | C | 200,00   | INR |
| 0000000001                                                             | 30 | 01.09.2025 | C | 200,00   | INR |
| 0000000002                                                             | 10 | 02.09.2025 | D | 600,00   | INR |
| 0000000002                                                             | 20 | 02.09.2025 | D | 400,00   | INR |
| 0000000003                                                             | 10 | 02.09.2025 | D | 1.000,00 | INR |
| 0000000003                                                             | 20 | 02.09.2025 | D | 800,00   | INR |
| 0000000003                                                             | 30 | 02.09.2025 | D | 200,00   | INR |
| 0000000004                                                             | 10 | 05.09.2025 | C | 2.500,00 | INR |
| 0000000004                                                             | 20 | 05.09.2025 | C | 500,00   | INR |
| 0000000005                                                             | 10 | 05.09.2025 | N | 2.500,00 | INR |
| 0000000005                                                             | 20 | 05.09.2025 | N | 800,00   | INR |
| 0000000005                                                             | 30 | 05.09.2025 | N | 700,00   | INR |

## Key Takeaways:

- FOR ALL ENTRIES reduces DB calls.
- Always check NOT INITIAL.
- Manual merging gives flexibility.
- Avoid SELECT inside LOOP.